Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE
(All Programs Accredited by NBA)
Department of Information Technology

Academic Year: 2025-26
Semester: VI
Class / Branch: TE-IT A
Subject: BI Lab
Name of Instructor: Prof. Apeksha Mohite

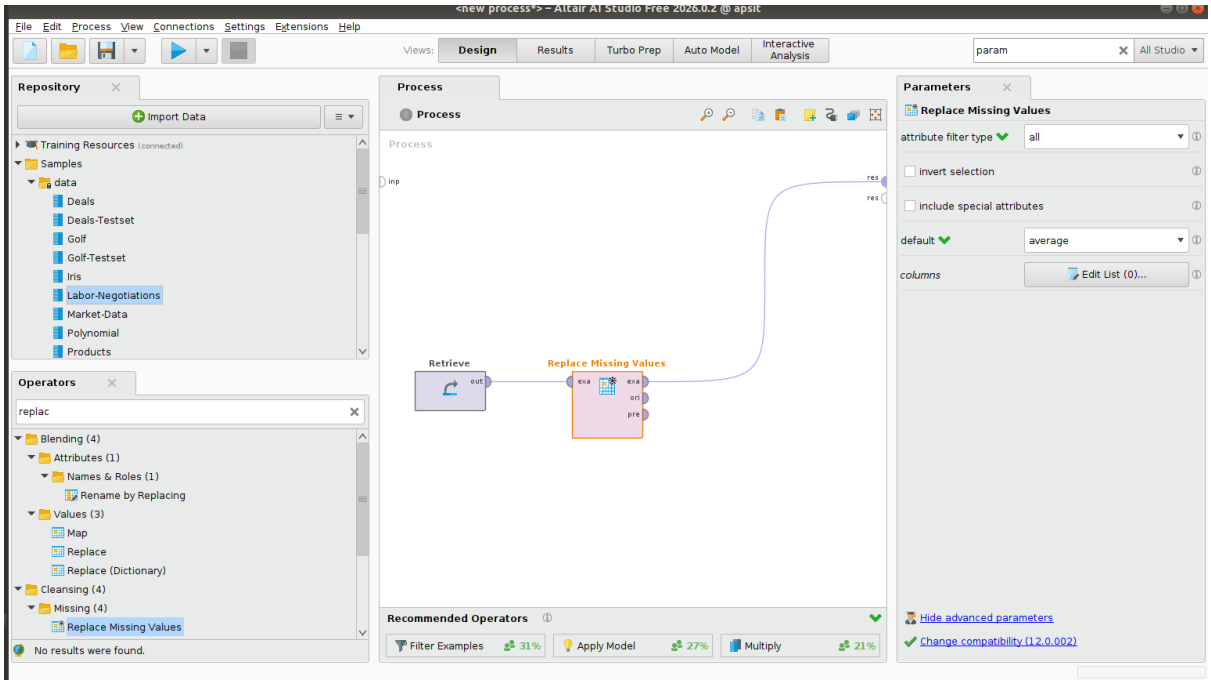
Name of Student:Anupras P. Davane
Student ID: 23104174
Date of Performance: 2026-02-16
Date of Submission: 2026-02-16

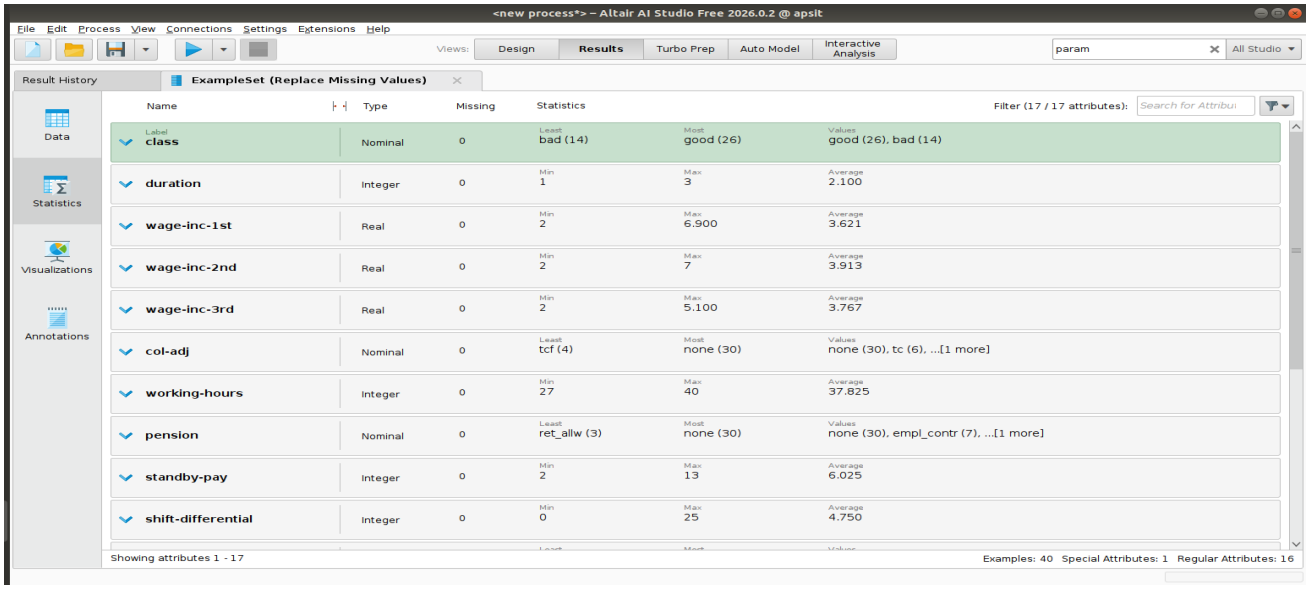
Experiment No. 05

Aim: To perform data pre-processing task and analyze results using open source tool, RapidMiner.

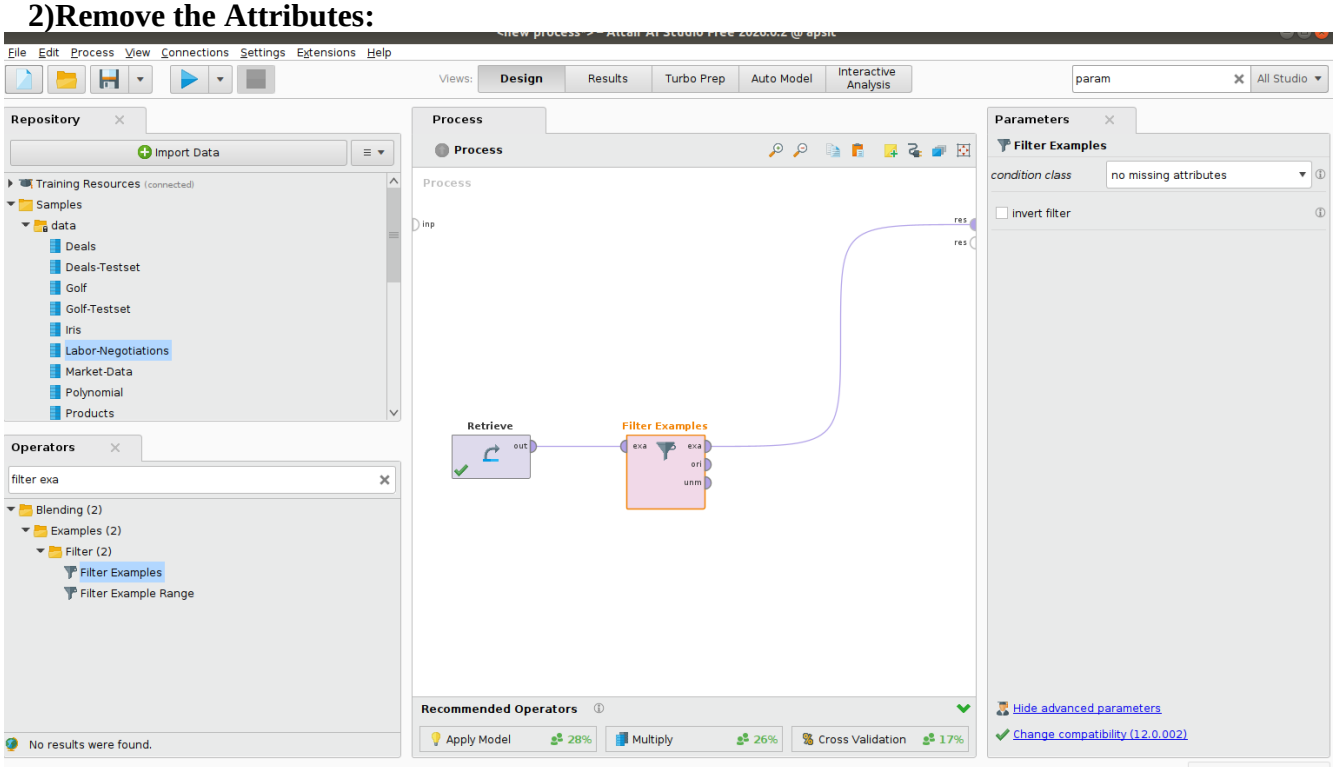
Output:

1)Replace the Missing Values:





2)Remove the Attributes:



Result History | ExampleSet (Filter Examples)

Data | Statistics | Visualizations | Annotations

Open in: Turbo Prep | Auto Model | Interactive Analysis

Filter (1 / 1 examples): all

Row No.	class	duration	wage-inc...	wage-inc...	wage-inc...	col-adj	working-...	pension	standby-...	shift-diff...	educatio...	statutor...	vacation
1	bad	3	2	2.500	2.100	tc	40	none	2	1	no	10	below-aver...

Result History | ExampleSet (Filter Examples)

Data | Statistics | Visualizations | Annotations

Filter (17 / 17 attributes): Search for Attribut

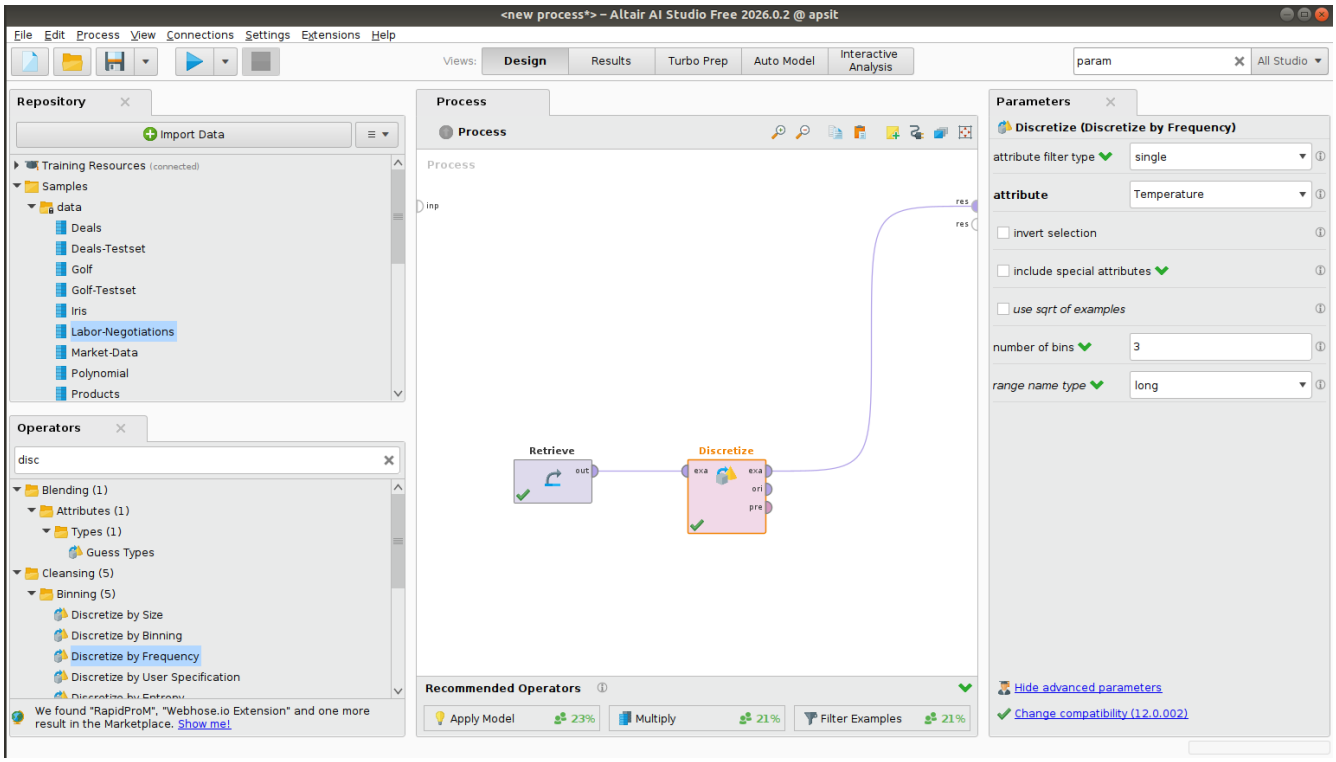
Name	Type	Missing	Statistics		
Label class	Nominal	0	Least good (0)	Most bad (1)	Values bad (1), good (0)
duration	Integer	0	Min 3	Max 3	Average 3
wage-inc-1st	Real	0	Min 2	Max 2	Average 2
wage-inc-2nd	Real	0	Min 2.500	Max 2.500	Average 2.500
wage-inc-3rd	Real	0	Min 2.100	Max 2.100	Average 2.100
col-adj	Nominal	0	Least tcf (0)	Most tc (1)	Values tc (1), none (0), ...[1 more]
working-hours	Integer	0	Min 40	Max 40	Average 40
pension	Nominal	0	Least ret_allw (0)	Most none (1)	Values none (1), empl_contr (0), ...[1 more]
standby-pay	Integer	0	Min 2	Max 2	Average 2
shift-differential	Integer	0	Min 1	Max 1	Average 1

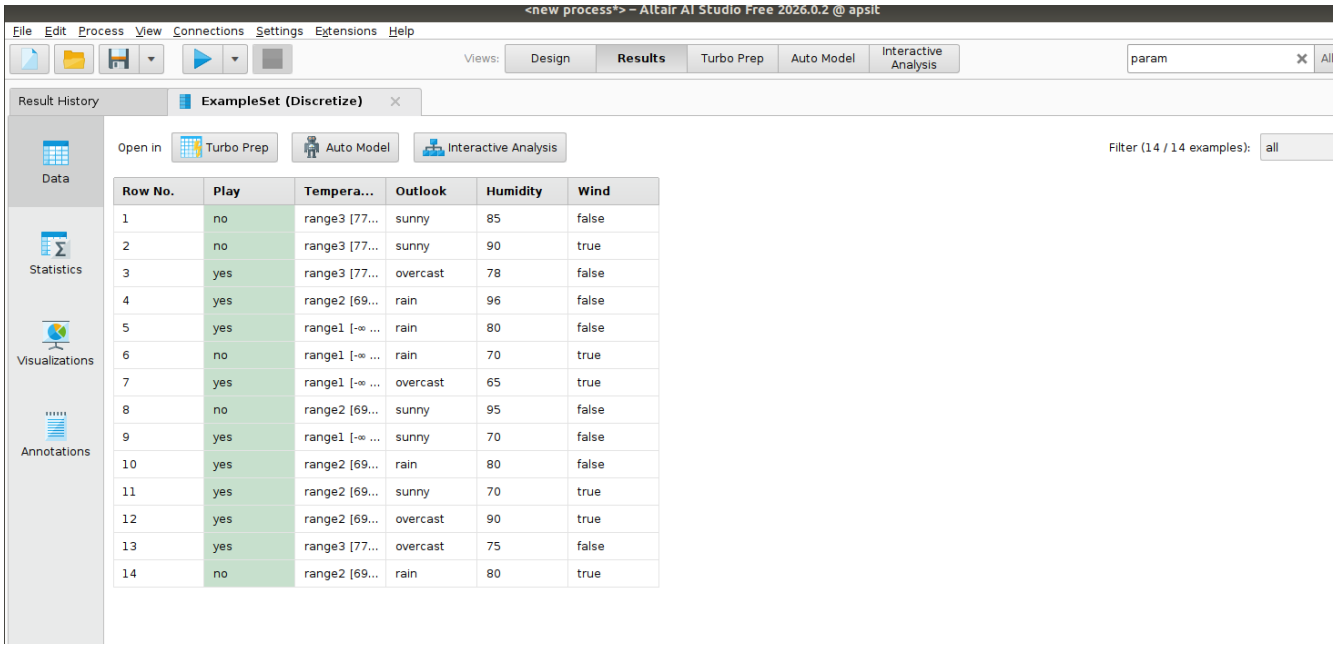
Showing attributes 1 - 17

Examples: 1 Special Attributes: 1 Regular Attributes: 16

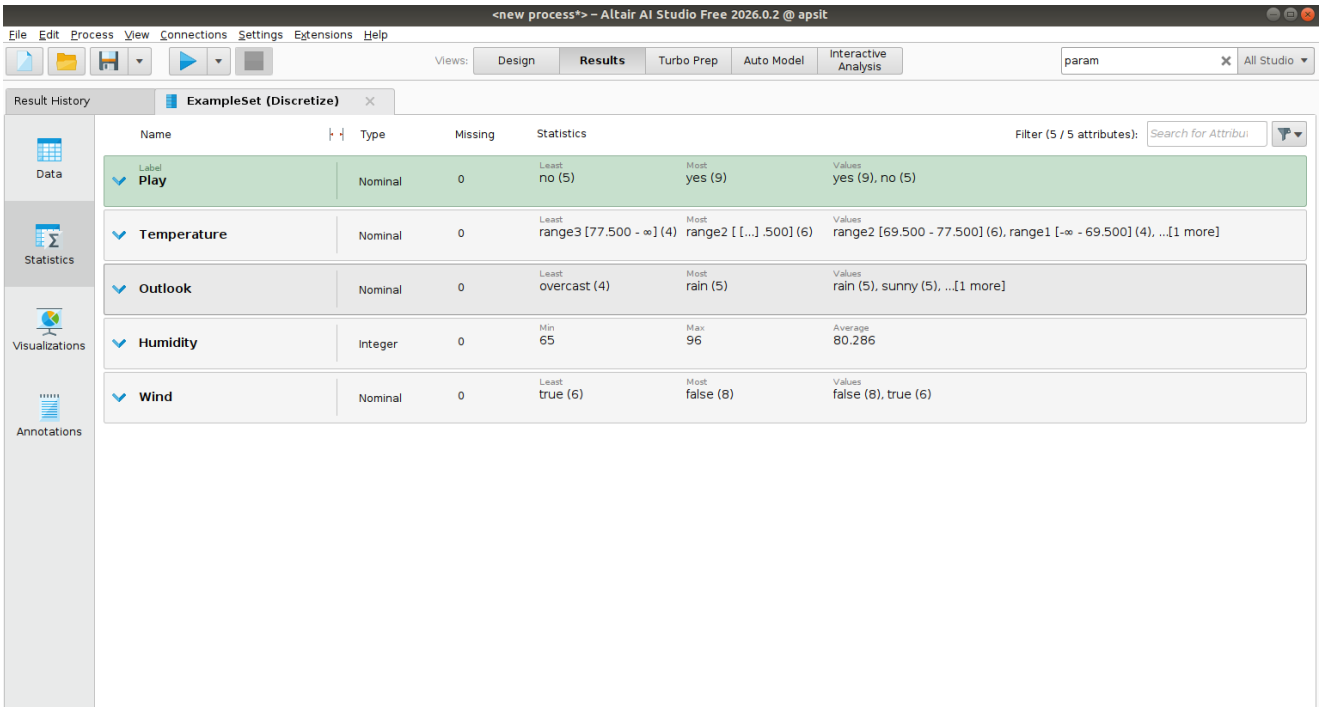
3)Discretize:

Golf Dataset

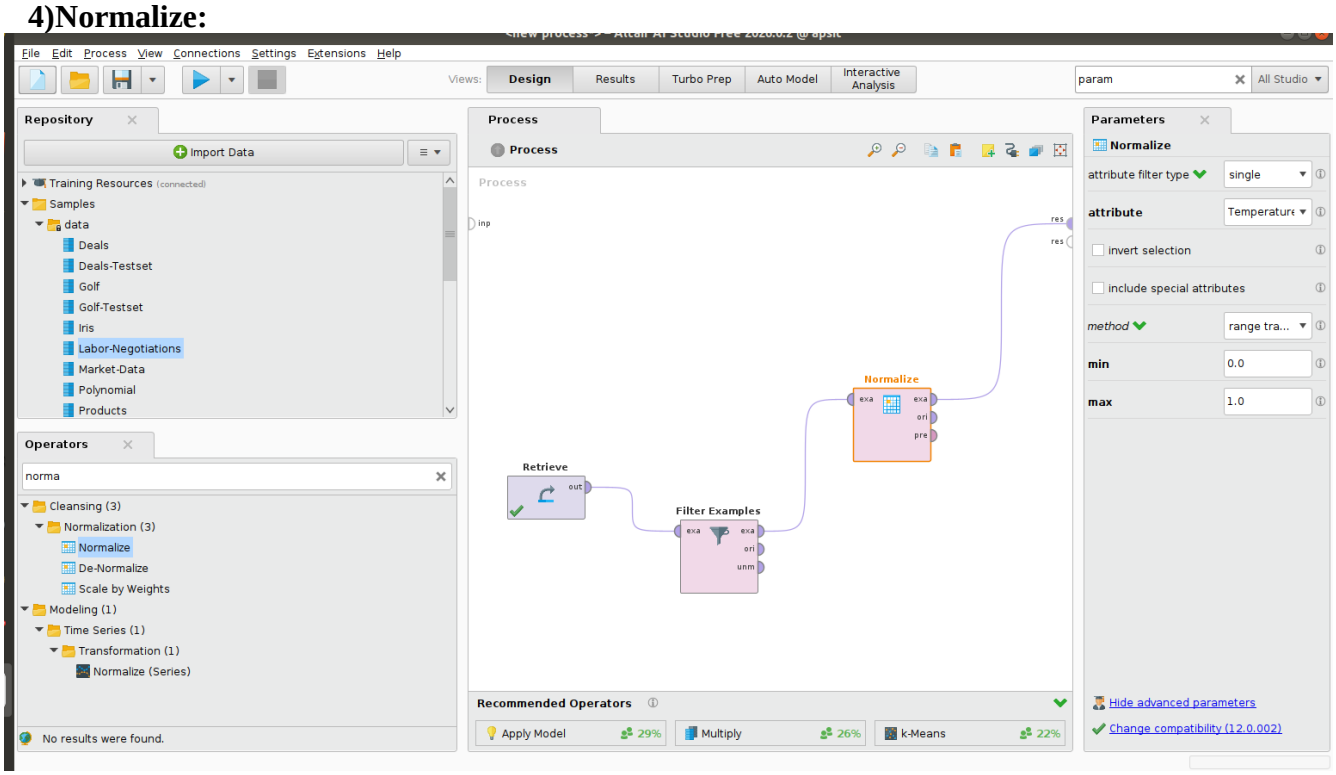


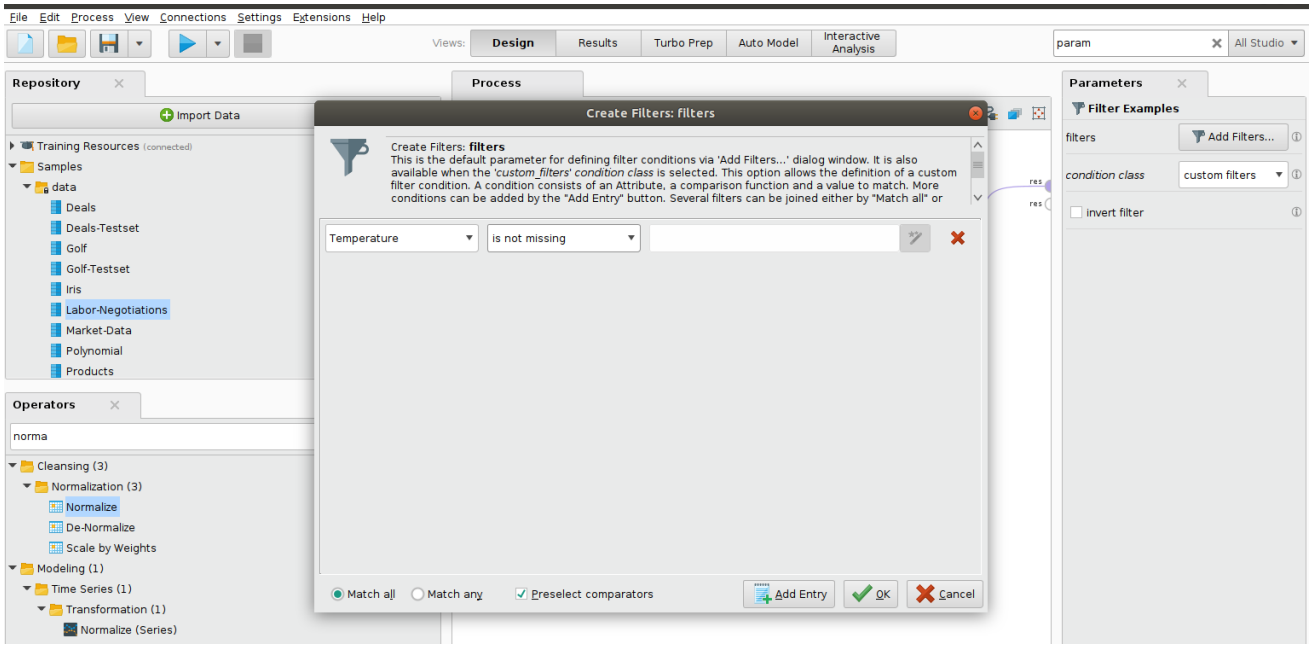


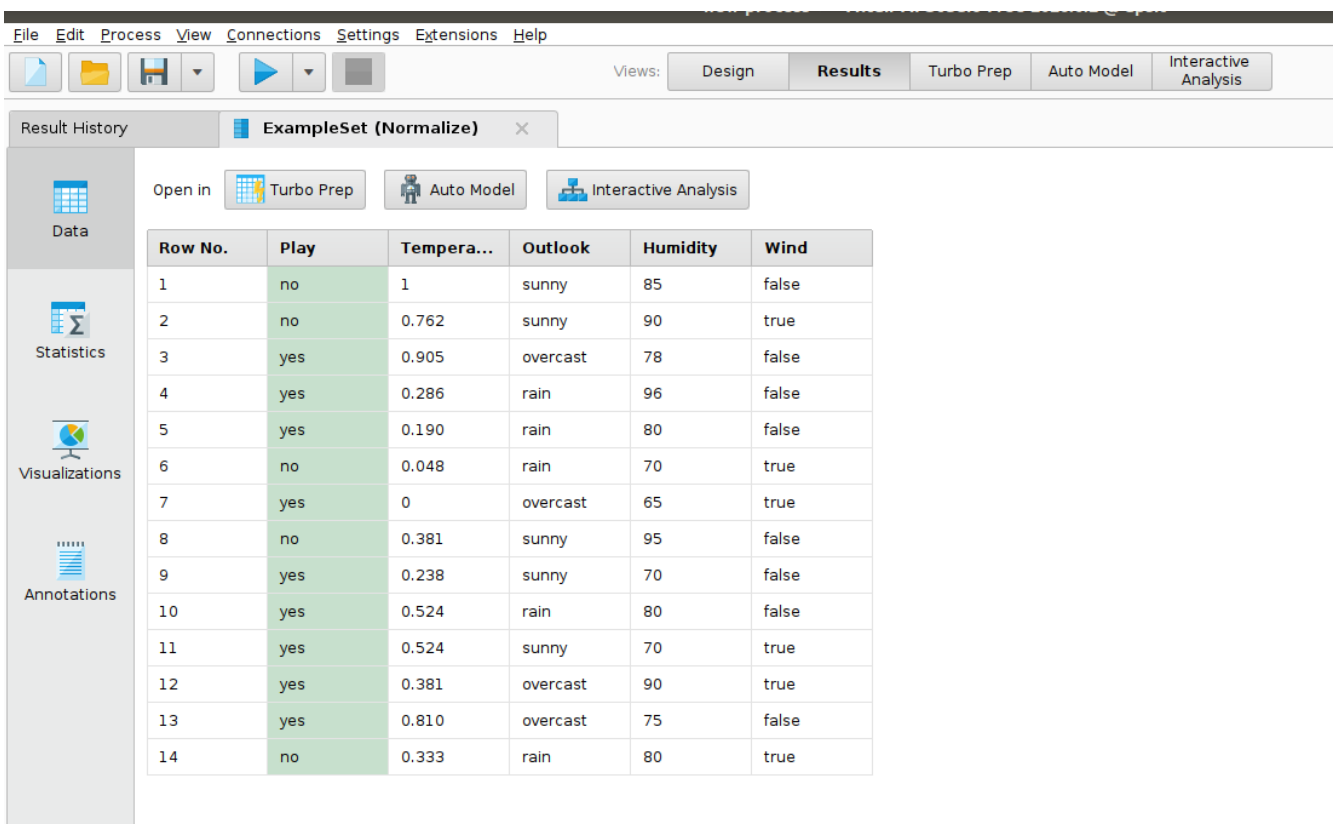
Row No.	Play	Tempera...	Outlook	Humidity	Wind
1	no	range3 [77...	sunny	85	false
2	no	range3 [77...	sunny	90	true
3	yes	range3 [77...	overcast	78	false
4	yes	range2 [69...	rain	96	false
5	yes	range1 [-∞ ...	rain	80	false
6	no	range1 [-∞ ...	rain	70	true
7	yes	range1 [-∞ ...	overcast	65	true
8	no	range2 [69...	sunny	95	false
9	yes	range1 [-∞ ...	sunny	70	false
10	yes	range2 [69...	rain	80	false
11	yes	range2 [69...	sunny	70	true
12	yes	range2 [69...	overcast	90	true
13	yes	range3 [77...	overcast	75	false
14	no	range2 [69...	rain	80	true


<new process*> – Altair AI Studio Free 2026.0.2 @ apsit
File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
param
All Studio
Result History
ExampleSet (Discretize)
Data
Statistics
Visualizations
Annotations
Name Type Missing Statistics
Filter (5 / 5 attributes):
Search for Attribut
Label Play Nominal 0 Least no (5) Most yes (9) Values yes (9), no (5)
Temperature Nominal 0 Least range3 [77.500 - ∞] (4) Most range2 [[...] .500] (6) Values range2 [69.500 - 77.500] (6), range1 [-∞ - 69.500] (4), ...[1 more]
Outlook Nominal 0 Least overcast (4) Most rain (5) Values rain (5), sunny (5), ...[1 more]
Humidity Integer 0 Min 65 Max 96 Average 80.286
Wind Nominal 0 Least true (6) Most false (8) Values false (8), true (6)

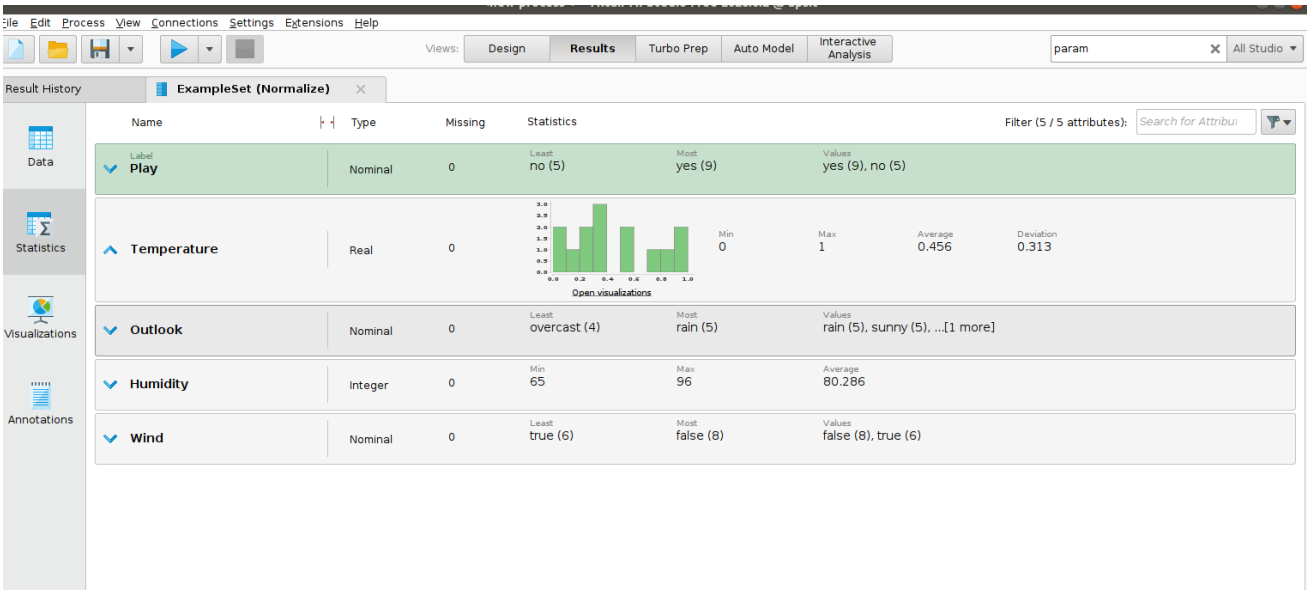
4)Normalize:


File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
param
All Studio
Repository
Import Data
Training Resources (connected)
Samples
data
Deals
Deals-Testset
Golf
Golf-Testset
Iris
Labor-Negotiations
Market-Data
Polynomial
Products
Operators
norma
Cleansing (3)
Normalization (3)
Normalize
De-Normalize
Scale by Weights
Modeling (1)
Time Series (1)
Transformation (1)
Normalize (Series)
No results were found.
Process
Retrieve
Filter Examples
Normalize
Recommended Operators
Apply Model 29%
Multiply 26%
k-Means 22%
Parameters
Normalize
attribute filter type single
attribute Temperature
invert selection
include special attributes
method range tra...
min 0.0
max 1.0
Hide advanced parameters
Change compatibility (12.0.002)

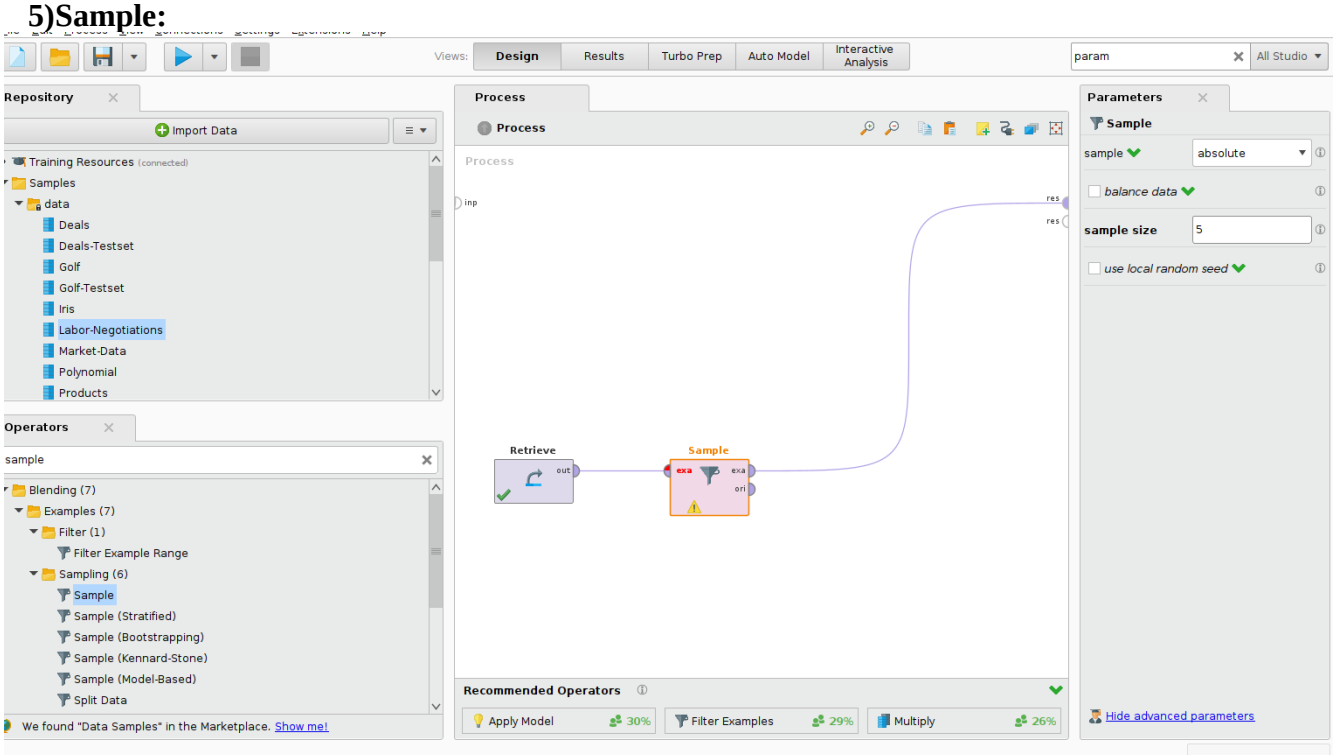


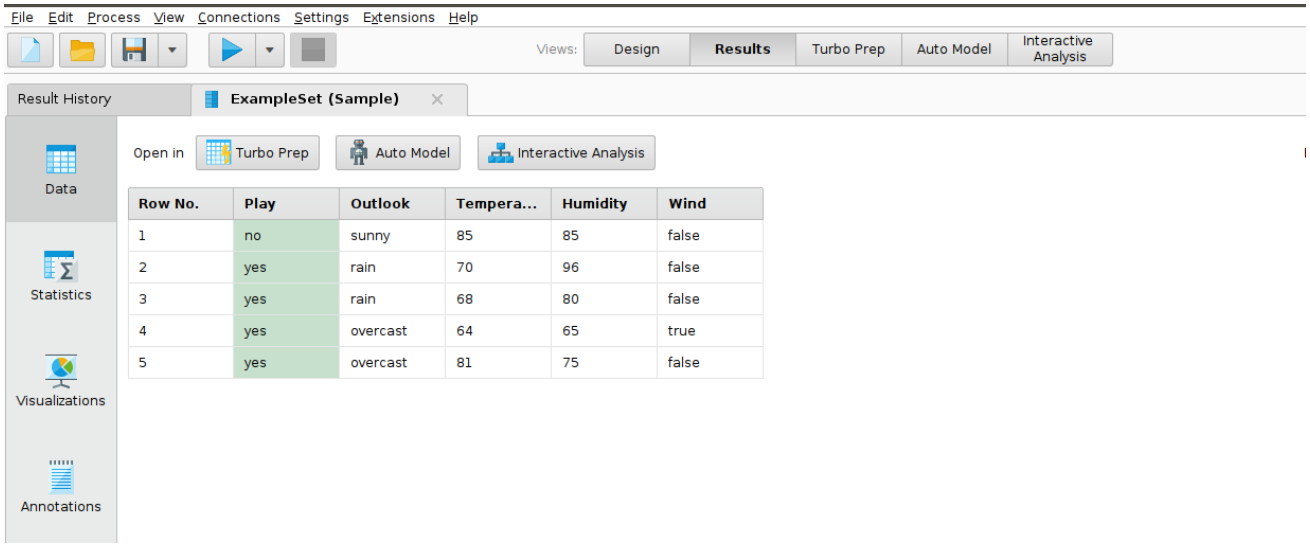


Row No.	Play	Tempera...	Outlook	Humidity	Wind
1	no	1	sunny	85	false
2	no	0.762	sunny	90	true
3	yes	0.905	overcast	78	false
4	yes	0.286	rain	96	false
5	yes	0.190	rain	80	false
6	no	0.048	rain	70	true
7	yes	0	overcast	65	true
8	no	0.381	sunny	95	false
9	yes	0.238	sunny	70	false
10	yes	0.524	rain	80	false
11	yes	0.524	sunny	70	true
12	yes	0.381	overcast	90	true
13	yes	0.810	overcast	75	false
14	no	0.333	rain	80	true


File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
param
All Studio
Result History
ExampleSet (Normalize)
Data
Statistics
Visualizations
Annotations
Name Type Missing Statistics
Filter (5 / 5 attributes):
Search for Attribut
Play Nominal 0 Least no (5) Most yes (9) Values yes (9), no (5)
Temperature Real 0 Min 0 Max 1 Average 0.456 Deviation 0.313
Open visualizations
Outlook Nominal 0 Least overcast (4) Most rain (5) Values rain (5), sunny (5), ...[1 more]
Humidity Integer 0 Min 65 Max 96 Average 80.286
Wind Nominal 0 Least true (6) Most false (8) Values false (8), true (6)

5)Sample:


Views: Design Results Turbo Prep Auto Model Interactive Analysis
param
All Studio
Repository
Import Data
Training Resources (connected)
Samples
data
Deals
Deals-Testset
Golf
Golf-Testset
Iris
Labor-Negotiations
Market-Data
Polynomial
Products
Operators
sample
Blending (7)
Examples (7)
Filter (1)
Filter Example Range
Sampling (6)
Sample
Sample (Stratified)
Sample (Bootstrapping)
Sample (Kennard-Stone)
Sample (Model-Based)
Split Data
We found "Data Samples" in the Marketplace. Show me!
Process
Process
Retrieve
Sample
Recommended Operators
Apply Model 30%
Filter Examples 29%
Multiply 26%
Parameters
Sample
sample absolute
balance data
sample size 5
use local random seed
Hide advanced parameters



Row No.	Play	Outlook	Tempera...	Humidity	Wind
1	no	sunny	85	85	false
2	yes	rain	70	96	false
3	yes	rain	68	80	false
4	yes	overcast	64	65	true
5	yes	overcast	81	75	false

Conclusion: In today's experiment, we learnt about data pre-processing using the tool called RapidMiner. We performed operations such as Replace the missing values, filtering (Removing attribute), Discretize, Normalize, and Sample. Using this tool we can see a complete visualization after running the operations in the form of actual table/relation as well as in statistical form.